

IR-Stepper-Control

The IR-Stepper-Control project is designed to operate safely and efficiently. It can be integrated into a smart parking system, industrial belt conveyors or automated people counter for retail stores.

This project's purposes are:

- Appropriate barrier movement – the motor starts working only in two cases.
 - 1) first one, if an object was detected by the IR sensor and the barrier position is at 0 degrees
 - 2) the barrier position is at 90 degrees while the IR sensor is in HIGH state (not detecting anything after the object passed by).
- Safe and efficient power system – for the stepper motor to receive power only when needed, it's driver is carefully controlled. This step required a relay, transistor, 1k resistor and one 1n4007 diode.
 - 1) The transistor is used as a switch, along with the resistor, which is used for safety reasons like limiting the current, is connected to an Arduino digital pin. Using an NPN transistor the logic is LOW-Side-Switching.
 - 2) As the pin is in HIGH state (the IR sensor pin also) the current flows to the relay coil, triggering the relay to switch to NO (Normally Open).
 - 3) The driver is supplied with power and the stepper motor turns.
- Current monitoring, event counter and system states:
 - 1) The driver's VCC wire is passed through the Hall sensor module's hole to measure current
 - 2) As the barrier goes up, the counter will be incremented. If the system is powered after being disconnected for a while a question will pop up on the display, “Doresti resetarea memoriei? (da/nu)”. If “da” the counter is automatically reset to zero, using the EEPROM memory, if “no” counting starts from the previous value it had,
 - 3) Depending on the current value one led will light up and a message will be displayed on the OLED. When a dangerous value is read, red led lights up, also the buzzer generates a short sound.

Project's components:

- 1) HW-131 power supply,
- 2) WCS1800 Hall current sensor module,
- 3) 28BYJ-48 stepper motor,
- 4) ULN2003 driver,
- 5) SRD-05VDC-SL-C relay,
- 6) IR infrared obstacle avoidance sensor module,
- 7) OLED display,
- 8) Active buzzer,
- 9) BD139 NPN transistor,
- 10) 1k ohm resistor,
- 11) 1n4007 diode,
- 12) 3 x 202 ohms resistor.